2. Network Topologies and Connections

Understanding network topologies

- This idea of creating a layout of a network is commonly referred to as a topology or **network topology.**
- It is used to help IT professionals understand how systems and devices are interconnected and to identify areas of improvement or where an issue may exist.
- It can be
  - Physical
  - Logical

Physical Topology

- It contains a high-level overview showing the actual arrangement and placement of devices within an organization.
- It also shows the types of network connections and connectors.

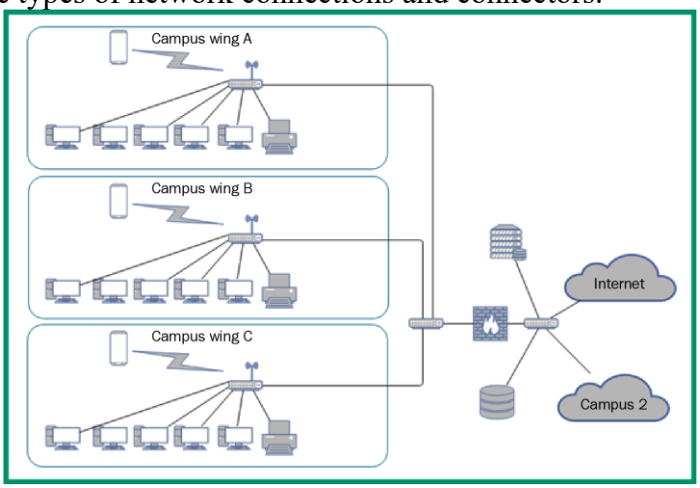


Fig. – Physical network topology

Logical Topology

- It focuses on the technical details of the network, such as the IP addresses and schemes, which demonstrate how data flows across the organization.
- It provides details about the different devices, such as switches, routers, and firewalls.

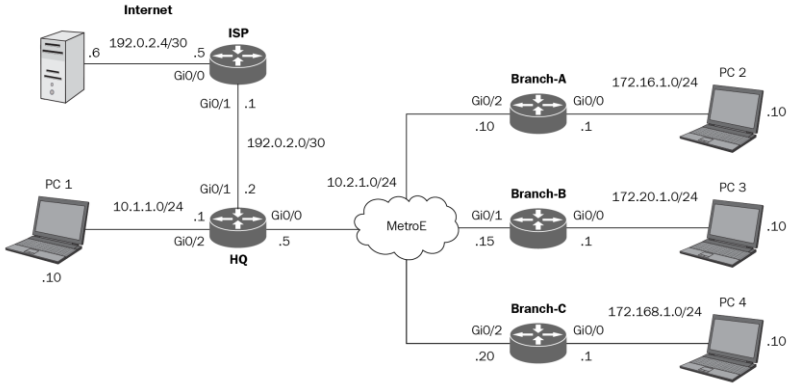


Fig. – Logical network topology

Network diagrams should always be updated whenever a change occurs on the network. A change can be anything such as an upgrade, implementing new technologies and devices, and replacing or even decommissioning a device.

Types of network topologies

There are various types of network topologies, and each has its advantages and disadvantages.

Bus

- This type of network topology was designed to use a single networking cable as the main backbone of the entire network.
- This single backbone allowed nodes such as computers to connect to and access any resource on the network.

The following diagram shows an example of a traditional bus topology:

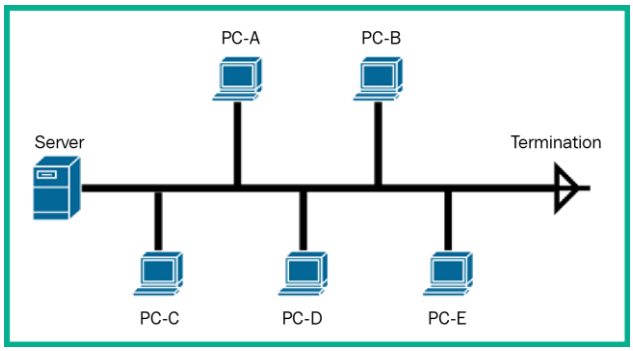


Fig. – Bus topology

- On each end of the backbone networking cable, some terminators are installed to terminate or ground the cable if there are any unwanted electrical signals on the wire.
- In **Shared Ethernet**, the message is sent to a specific destination **Media Access Control** (**MAC**) address.
- Each node receives the message but only the **Network Interface Card** (**NIC**) whose MAC address matches the destination MAC address in the destination section of the Ethernet frame will accept the message (packet).
- All other NICs will discard this message (packet).

The following diagram shows an example of a host sending a Shared Ethernet message:

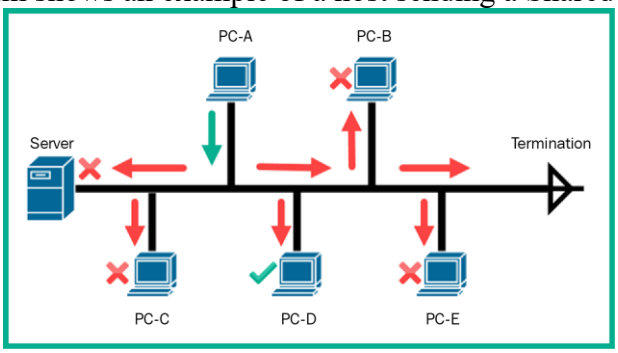


Fig. – Shared Ethernet on a bus topology

Advantages

- It is simple to design.
- It requires less cabling.
- Cost is less.

Disadvantages

- Only one device can send a message at a time.
- If multiple devices broadcast their message, network collisions can occur.
- There is no redundant path (alternate path) in case of a network failure.
- If there's a break at any point along the backbone network cable, this will result in an entire network outage.
- The bus topology does not support scalability.
- Data is not secure as it is broadcasted and every host receives it.

Ring

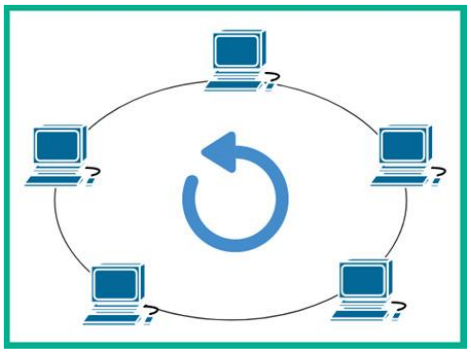


Fig. – Ring topology

- Each host (device) is interconnected to another host to create the ring topology.
- Here communication only occurs in a single direction.

Advantages

- Predictable/control communication in traditional implementation

Disadvantages

- The message from the sender is passed to each host on the network until it arrives at the destination host.
- Only one host can send a message at a time on the network.
- If one node is not available, the entire network goes down.

Dual ring

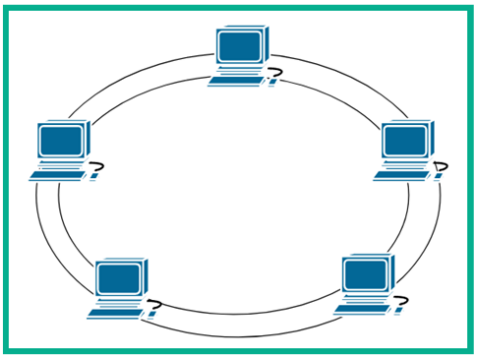


Fig. – Dual ring topology

- It has two logical ring connections on each node on the network.
- The dual ring allows communication to occur in opposite directions on each ring.
- Here, one ring will allow traffic to flow clockwise only, and counterclockwise in the other ring.

Star

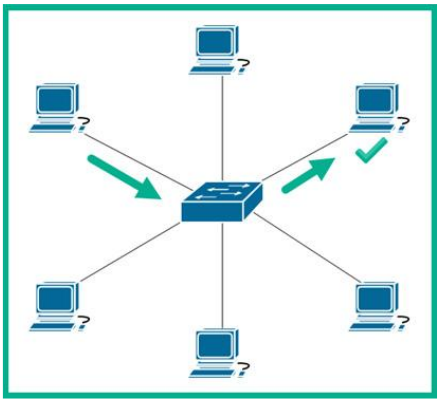


Fig. – Star topology

- It allows the hosts to connect to a single networking device such as a switch/hub.
- If any host wants to send a message to another host, the sender will forward the message to the network switch, which forwards the message to the destination host.
- The switch functions as the network intermediary device to forward messages between a sender and receiver on the network.

Advantages

- If one host on the network is offline or unavailable, the entire network is not affected. The hosts can continue to send and receive messages simultaneously.
- It supports scalability.

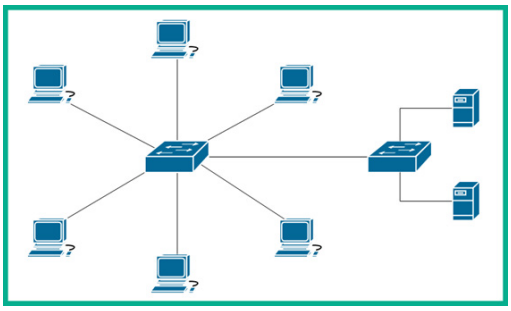


Fig. – Scalability in a star topology

- More than one host can send messages at a time by using switch.
- Data is secured as it is received by the intended receiver only.

Disadvantages

- If the central networking device such as the switch goes down or becomes unavailable, the entire network goes down.
- There is no redundant or alternate path in case of network failure.

The following diagram shows a visual representation of a network outage in a star topology:

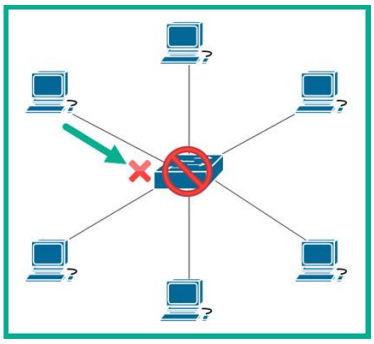


Fig. – Network outage

Mesh

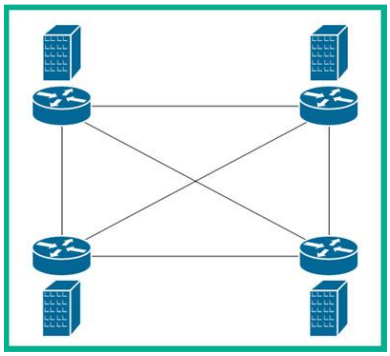


Fig. – Mesh topology

- Each device within the network is connected to all other devices, creating a physical and logical mesh design.

Advantages

- It creates multiple and redundant paths within the network. So, if one path goes down or becomes unavailable, there are multiple other paths available to forward network traffic between a source and a destination.
- It provides full redundancy for traffic flow between any source and any destination as all devices are interconnected.
- Data is secured as it is received by the intended recipient only.

Disadvantages

- It can be complex to troubleshoot if an issue occurs due to the number of available paths.
- It becomes quite costly and complex as the network increases in size.

No. of links = N(N-1)/2, where N is the no. of devices
e.g., if there are 5 devices, the total number of links = 5(5-1)/2 = 10.
Adding just 1 additional device no. of links =6(6-1)/2 = 15.

Hybrid

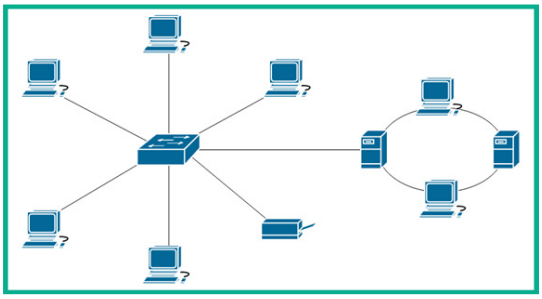


Fig. – Hybrid topology

- Itis a combination of two or more different topologies that are interconnected to create a unified network.
- The above network consists of both star and ring topologies.
- Often, network professionals use this topology within their organizations to interconnect end devices such as computers and servers, as well as interconnect their branch offices together to share resources.

Hub and Spoke

- It is designed to provide a centralized node that acts as the main hub for all other nodes to interconnect.
- Each node is connected to the hub and these connections are referred to as spokes.
- It does not allow nodes to directly connect; all network traffic is sent through the hub which is responsible for forwarding traffic to another node on the network.

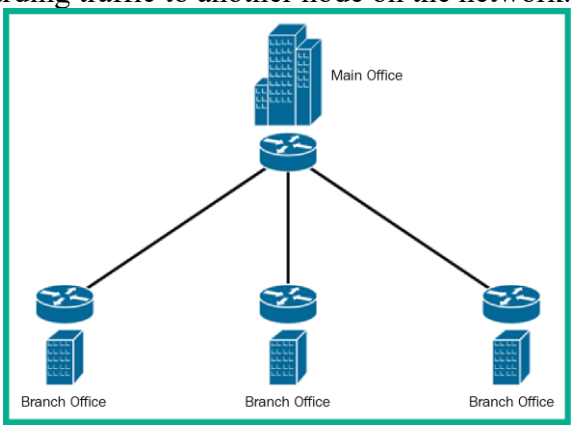


Fig. – Hub and spoke topology

- As shown in the preceding diagram, the organization has one main office that will function as the headquarters that contains all the servers, while three remote branch offices will be used to provide services to customers are various geographic locations.
- Whenever a user or device needs to access a resource at another branch location, the traffic is sent to the main officer router (hub), which then forwards the traffic to the destination network.

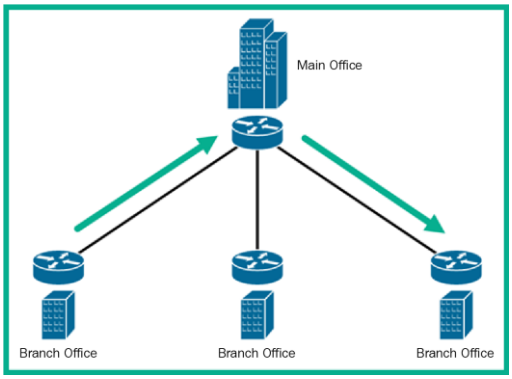


Fig. – Communication over a hub and spoke network

- This topology is commonly used by medium-sized and large organizations to interconnect their remote branch offices to their main office location.
- It is a cost-efficient design that allows organizations to save on internet subscription fees and improve their security monitoring.

Discovering network types

Network types are used to define the geographic boundary or limitation of a network and help network professionals understand the relationship between the connected devices within the network.

Peer-to-peer

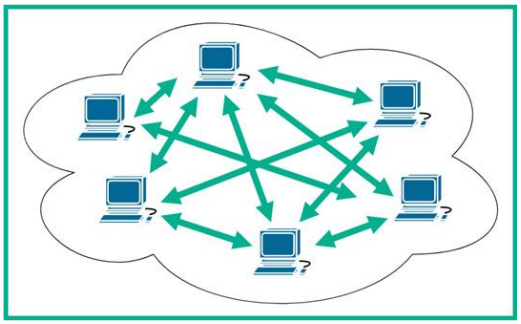


Fig. – Peer-to-peer model

- Here each client is logically connected and shares their resources with other clients on the network without the need for a centralized, dedicated server to provide the resources.
- If a client has a printer, that client can be configured to share the printer as a resource with other clients on the network. The client that is sharing the printer or resource becomes a server, while the others host (computers) are clients that are accessing or requesting the resource.
- It does not support scalability well.
- These are typically implemented as workgroups in many small organizations.
- There's no centralized management of resources.
- Security is a huge concern as each node is responsible for managing its security posture.
- A real-world example of a peer-to-peer network is sharing files using a *torrent*.

Client-server

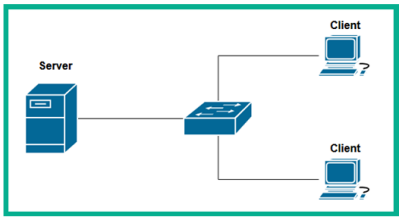


Fig. – Client-server model

- It contains dedicated devices that provide a resource or service to hosts on a network. These devices are referred to as servers.
- A server can be any device running an application that allows other hosts on the network to access the resources stored or being provided by the server.
- The server can be configured as a file server to centrally store files and data for other devices and users on the network. The clients on the network are the devices that are requesting a service or resource.
- Scalability allows the network to grow while ensuring the resources are centrally accessible by all clients on the network.

Personal area network

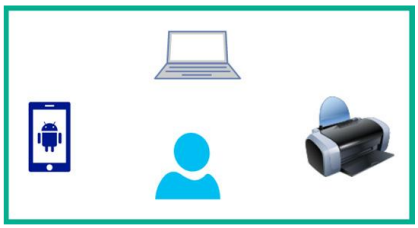


Fig. – Personal area network

- It is a small network that is usually created by a user to interconnect their devices, such as a laptop, smartphone, wireless headphones, or a wireless printer.
- A user can set up a wired or wireless network to ensure their devices are connected to the same network.
- It is not designed to connect to a larger network with other devices.
- A real-world example of a PAN is connecting a laptop and smartphone to a wireless printer at home to print pictures.
- Each device is responsible for managing its security.

Local area network

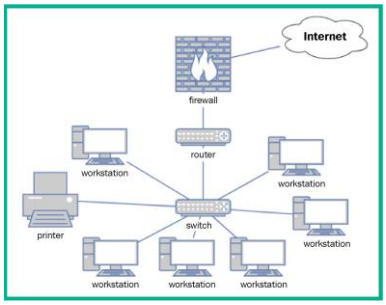


Fig. – LAN network model

- It exists within a single geographic location, such as within a building.
- A LAN allows all devices within an office location or building to be interconnected and share resources.
- Interconnecting all the devices, such as the computers, servers, and network printers, into a single physical network creates a LAN that allows everyone and all devices on the same network to easily share resources.
- As shown in the preceding diagram, the network contains a few workstations that are used by the employees within a small office, and each end device, such as a computer and printer, is connected to the network switch.
- The switch connects to the router, which functions as the **default gateway** to forward traffic from the internal network to a remote/foreign network such as the internet.
- The router connects to the firewall to filter inbound and outbound traffic between the internal network and the internet.

Wide area network

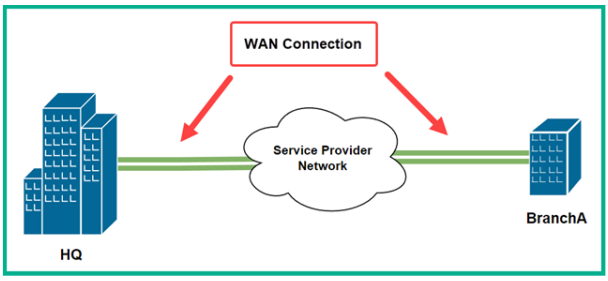


Fig. – WAN

- It allows organizations to extend their LAN and its resources over a large geographic distance.
- As time goes by, the organization will grow and create new remote office locations within other cities or countries to support its customers.
- The organization may have all its servers located at its main office and want to share access with employees who are working at a remote office.
- Using a WAN allows the organizations to extend their LAN from their main office over a large geographic distance to each of the remote offices as needed.
- The WAN connection is established by a **Managed Service Provider** (**MSP**) or an **Internet Service Provider** (**ISP**).
- All traffic sent into a WAN connection by an organization remains private as the traffic passes through the service provider's network infrastructure.

Metropolitan area network

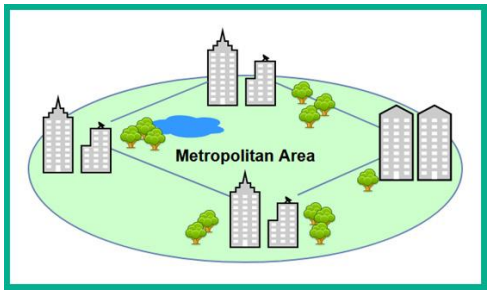


Fig. – Metropolitan area network type

- In large cities, some organizations have multiple branch offices within the same city and need interconnectivity between their offices to share resources with employees.
- Using a **metropolitan area network** (**MAN**), a service provider can interconnect all the branch offices of a single company within the same city.
- Some cities offer a MAN to residences as a low-cost, high-speed connection to the internet. These MANs can be used to incentivize residents to move to that city.

Wireless local area network

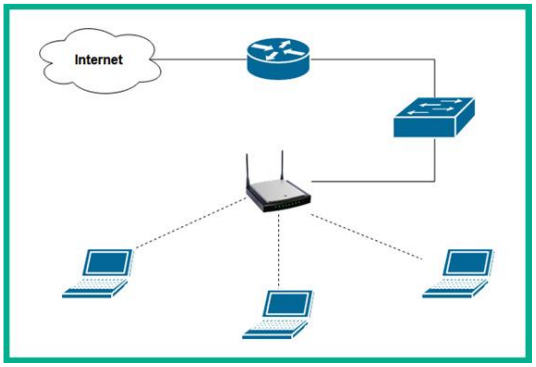


Fig. – WLAN network type

- It is a wireless local area network that has one or more Access Points or wireless routers to allow wireless capable clients to connect and share resources.
- APs are connected to a network switch that provides wireless connectivity to wireless clients, as shown in the above diagram.
- The wireless router generates a wireless signal using radio frequencies on the 2.4 GHz, 5 GHz, and 6GHz bands based on the **Institute of Electrical and Electronics Engineers** (**IEEE**) 802.11 standard.
- The wireless network generated by the wireless router is on a unique **Internet Protocol** (**IP**) network, while the wireless router is connected to a wired network on a different IP network.
- Hence, a wireless router performs routing between the two different IP networks, from wireless to wired, and between the different IP networks.

Important note

Wireless routers are commonly used on small networks such as within homes and small businesses. An **Access Point** (**AP**) is a layer 2 device that provides a wireless network to allow wireless clients to establish a connection and access the resources on the wired network.